

Make a SQL database of a game

1. Create a database called GameDB.

Answer:

```
CREATE DATABASE GameDB;
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0023 seconds.)

```
CREATE DATABASE CricketerStatsDB;
```

2. Create an table called `item_types` with the following item types:

- melee
- ranged
- defense
- potion
- magic

Answer:

```
CREATE TABLE item_types (  
    name VARCHAR(50) PRIMARY KEY  
);
```

```
INSERT INTO item_types (name) VALUES  
( 'melee' ),  
( 'ranged' ),  
( 'defense' ),  
( 'potion' ),  
( 'magic' );
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0310 seconds.)

```
CREATE TABLE item_types ( name VARCHAR(50) PRIMARY KEY );
```

[\[Edit inline \]](#) [\[Edit query \]](#)

✓ 5 rows inserted. (Query took 0.0120 seconds.)

```
INSERT INTO item_types (name) VALUES ( 'melee' ), ( 'ranged' ), ( 'defense' ), ( 'potion' ), ( 'magic' );
```

[\[Edit inline \]](#) [\[Edit query \]](#)

3. Create a table `items` using `name` as the primary key. Each item has:

- A unique name
- Durability (default 100)
- Type (from `item_type_enum`)

Answer:

```
CREATE TABLE items (  
    name VARCHAR(50) PRIMARY KEY,  
    durability INT DEFAULT 100,  
    item_type VARCHAR(50) NOT NULL  
    FOREIGN KEY (item_type) REFERENCES item_types(name)  
);
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0135 seconds.)

```
CREATE TABLE items ( name VARCHAR(50) PRIMARY KEY, durability INT DEFAULT 100, item_type VARCHAR(50) NOT NULL,  
FOREIGN KEY (item_type) REFERENCES item_types(name) );
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

4. Create a table `players` with the following:

- `id` (Primary Key)
- `name`
- `health` (default 100)
- `equipped_item_name` (Foreign key referencing `items(name)`)

Answer:

```
CREATE TABLE players (  
    id SERIAL PRIMARY KEY,  
    name VARCHAR(50) NOT NULL,  
    health INT DEFAULT 100,  
    equipped_item_name VARCHAR(50),  
    FOREIGN KEY (equipped_item_name) REFERENCES items(name)  
);
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0137 seconds.)

```
CREATE TABLE players ( id SERIAL PRIMARY KEY, name VARCHAR(50) NOT NULL, health INT DEFAULT 100,  
equipped_item_name VARCHAR(50), FOREIGN KEY (equipped_item_name) REFERENCES items(name) );
```

[\[Edit inline \]](#) [\[Edit \]](#)

5. Create a table `inventory` with:

- `id` (Primary Key)
- `player_id` (Foreign key)

- item_name (Foreign key referencing items(name))
- amount (default 1)

Answer:

```
CREATE TABLE inventory (
  id SERIAL PRIMARY KEY,
  player_id BIGINT UNSIGNED NOT NULL,
  item_name VARCHAR(50) NOT NULL,
  amount INT DEFAULT 1,
  FOREIGN KEY (player_id) REFERENCES players(id) ON DELETE CASCADE,
  FOREIGN KEY (item_name) REFERENCES items(name) ON DELETE CASCADE
);
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0133 seconds.)

```
CREATE TABLE inventory ( id SERIAL PRIMARY KEY, player_id BIGINT UNSIGNED NOT NULL, item_name VARCHAR(50) NOT
NULL, amount INT DEFAULT 1, FOREIGN KEY (player_id) REFERENCES players(id) ON DELETE CASCADE, FOREIGN KEY
(item_name) REFERENCES items(name) ON DELETE CASCADE );
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

6. Insert data into items.

name	durability	item_type
Fire Bow	80	ranged
Iron Sword	100	melee
Magic Wand	60	magic
Shield	120	defense
Healing Potion	1	potion
Gun	90	ranged

Answer:

```
INSERT INTO items (name, durability, item_type) VALUES
('Fire Bow', 80, 'ranged'), ('Iron Sword', 100,
'melee'),
('Magic Wand', 60, 'magic'),
('Shield', 120, 'defense'),
('Healing Potion', 1, 'potion'),
('Gun', 90, 'ranged');
```

✓ 6 rows inserted. (Query took 0.0072 seconds.)

```
INSERT INTO items (name, durability, item_type) VALUES ('Fire Bow', 80, 'ranged'), ('Iron Sword', 100, 'melee'),
('Magic Wand', 60, 'magic'), ('Shield', 120, 'defense'), ('Healing Potion', 1, 'potion'), ('Gun', 90, 'ranged');
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

7. Insert data into players.

name	health	equipped_item
Archer Alice	85	Fire Bow
Knight Karl	100	Iron Sword
Mage Mira	70	Magic Wand
Tank Tom	120	Shield

Answer:

```
INSERT INTO players (name, health, equipped_item_name) VALUES  
( 'Archer Alice', 85, 'Fire Bow'), ( 'Knight Karl', 100, 'Iron  
Sword'),  
( 'Mage Mira', 70, 'Magic Wand'),  
( 'Tank Tom', 120, 'Shield');
```

✔ 4 rows inserted.

Inserted row id: 4 (Query took 0.0039 seconds.)

```
INSERT INTO players (name, health, equipped_item_name) VALUES ( 'Archer Alice', 85, 'Fire Bow'), ( 'Knight Karl',  
100, 'Iron Sword'), ( 'Mage Mira', 70, 'Magic Wand'), ( 'Tank Tom', 120, 'Shield');
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

8. Insert data into inventory.

player_name	item_name	amount
Archer Alice	Fire Bow	1
Archer Alice	Healing Potion	3
Knight Karl	Iron Sword	1
Knight Karl	Healing Potion	1
Mage Mira	Magic Wand	1
Mage Mira	Healing Potion	2
Tank Tom	Shield	1
Tank Tom	Healing Potion	5
Tank Tom	Gun	1

Answer:

```
INSERT INTO inventory (player_id, item_name, amount) VALUES  
(1, 'Fire Bow', 1),  
(1, 'Healing Potion', 3),  
(2, 'Iron Sword', 1),  
(2, 'Healing Potion', 1),
```

```
(3, 'Magic Wand', 1),
(3, 'Healing Potion', 2),
(4, 'Shield', 1),
(4, 'Healing Potion', 5),
(4, 'Gun', 1);
```

✓ 9 rows inserted.

Inserted row id: 9 (Query took 0.0036 seconds.)

```
INSERT INTO inventory (player_id, item_name, amount) VALUES (1, 'Fire Bow', 1), (1, 'Healing Potion', 3), (2,
'Iron Sword', 1), (2, 'Healing Potion', 1), (3, 'Magic Wand', 1), (3, 'Healing Potion', 2), (4, 'Shield', 1), (4,
'Healing Potion', 5), (4, 'Gun', 1);
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

Queries

a) Display all players.

Answer:

```
SELECT * FROM players;
```

id	name	health	equipped_item_name
1	Archer Alice	85	Fire Bow
2	Knight Karl	100	Iron Sword
3	Mage Mira	70	Magic Wand
4	Tank Tom	120	Shield

b) Show all item names and their types.

Answer:

```
SELECT name, item_type FROM items;
```

name	item_type
Shield	defense
Magic Wand	magic
Iron Sword	melee
Healing Potion	potion
Fire Bow	ranged
Gun	ranged

c) Show all players with health greater than 90.

Answer:

```
SELECT name, health FROM players WHERE health > 90;
```

name	health
Knight Karl	100
Tank Tom	120

d) Show the inventory of each player.

Answer:

```
SELECT p.name AS player_name, i.name AS item_name, inv.amount
FROM inventory inv
JOIN players p ON inv.player_id = p.id
JOIN items i ON inv.item_name = i.name;
```

player_name	item_name	amount
Tank Tom	Shield	1
Mage Mira	Magic Wand	1
Knight Karl	Iron Sword	1
Archer Alice	Healing Potion	3
Knight Karl	Healing Potion	1
Mage Mira	Healing Potion	2
Tank Tom	Healing Potion	5
Archer Alice	Fire Bow	1
Tank Tom	Gun	1

e) Count the number of items in each player's inventory.

Answer:

```
SELECT p.name, COUNT(*) AS item_count
FROM inventory inv
JOIN players p ON inv.player_id = p.id
GROUP BY p.name;
```

player_name	item_name	amount
Tank Tom	Shield	1
Mage Mira	Magic Wand	1
Knight Karl	Iron Sword	1
Archer Alice	Healing Potion	3
Knight Karl	Healing Potion	1
Mage Mira	Healing Potion	2
Tank Tom	Healing Potion	5
Archer Alice	Fire Bow	1
Tank Tom	Gun	1

f) Find players who have more than 2 potions.

Answer:

```
SELECT p.name
FROM inventory inv
JOIN players p ON inv.player_id = p.id
JOIN items i ON inv.item_name = i.name
WHERE i.item_type = 'potion' AND inv.amount > 2;
```

name
Archer Alice
Tank Tom

g) Show all items of type 'ranged'.

Answer:

```
SELECT * FROM items WHERE item_type = 'ranged';
```

name	durability	item_type
Fire Bow	80	ranged
Gun	90	ranged

h) Find the total number of items each player owns (sum of amounts).

Answer:

```
SELECT p.name, SUM(inv.amount) AS total_items
FROM inventory inv
JOIN players p ON inv.player_id = p.id
GROUP BY p.name;
```

name	total_items
Archer Alice	4
Knight Karl	2
Mage Mira	3
Tank Tom	7

i) Find the most durable item.

Answer:

```
SELECT name, durability FROM items
ORDER BY durability DESC
LIMIT 1;
```

name	durability
Shield	120

j) Find players who have more than one type of item.

Answer:

```
SELECT p.name
FROM inventory inv
JOIN players p ON inv.player_id = p.id
GROUP BY p.name
HAVING COUNT(DISTINCT inv.item_name) > 1;
```

name
Archer Alice
Knight Karl
Mage Mira
Tank Tom
